

46-923, Fall 2020: Homework #1

Due 12:01 PM, Monday, November 9

All of this work should be done within a Jupyter notebook. Please format it consistently with the notebooks that we used during Financial Data Science I. In particular, place emphasis on making your comments, code, and plots readable.

Turn in both the .ipynb file and a .pdf version.

1. Return to Question Two of Homework 5 from Financial Data Science I. Refit the model with all of the predictors, but first transform each predictor on the ratio scale using the Yeo-Johnson transformation. (The objective here is to give each predictor distribution more of a “normal” shape.)

Then, refit the model, but use the Box-Cox procedure to find the optimal transformation of the response.

Show the results of the resulting model fit, including the plot of residuals versus fitted values.

2. Create a new `treg()` function like the one demonstrated in lecture, but now include the number of degrees of freedom for the error distribution as an additional parameter to be estimated via maximum likelihood.

Print the function that you wrote and attach it to your homework, and also demonstrate that it works by using a few simulations.

The function should also return the covariance matrix for the MLE.

Comment: There is nothing wrong with having a non-integer valued number of degrees of freedom with the t -distribution.

3. Let $X_1, X_2, \dots, X_n$ be i.i.d. from the $\text{Beta}(\alpha, \beta)$ distribution. Find **numerically** the maximum likelihood estimators for α and β . Mimic the function `gammamle()` that we created in lecture.

The function should also return the covariance matrix for the MLE.

Print the function that you wrote and attach it to your homework, and also demonstrate that it works by using a few simulations.